

Assignment 5 Report

Part 1: Database Creation

1. create table Department (DID integer, Dname varchar(100), DLocation varchar(500),
constraint pk_dept primary key (DID));
create table Employee (EID integer, Ename varchar(100), Eage integer, Edeptid integer,
constraint pk_empl primary key (EID), foreign key (Edeptid) references
Department(DID));
2.
 - a. Primary keys:
 - i. constraint pk_dept primary key (DID)
 - ii. constraint pk_empl primary key (EID)
 - b. Foreign key:
 - i. foreign key (Edeptid) references Department(DID)
3. Eage integer check Eage between 18 and 99
4. We insert into Department first because otherwise there will be a foreign key constraint violation if we try inserting into Employee first (there needs to be an associated department id for each new Employee entry which would not be possible without having valid department ids found inside Department)
insert into Department values (1, 'Human Resources', '5th Floor');
insert into Department values (2, 'DevOps', '5th Floor');
insert into Department values (3, 'C-Suite', '4th Floor');
insert into Department values (4, 'Janitorial Staff', '6th Floor');

insert into Employee values (1, 'Sachin Tendulkar', 65, 1);
insert into Employee values (2, 'Virat Kohli', 45, 2);
insert into Employee values (3, 'Jasprit Bumrah', 25, 2);
insert into Employee values (4, 'Rohit Sharma', 30, 3);
insert into Employee values (5, 'Ravindra Jadeja', 29, 1);

commit;

5. When I try to insert the following entry into Employee with an invalid foreign key (DID) value of 0, I get this result:

```
insert into Employee values (0, 'Sachin Tendulkar', 50, 0)
Error report -
ORA-02291: integrity constraint (APETER40.ED_ID) violated -
```

<https://docs.oracle.com/error-help/db/ora-02291/>

More Details :

<https://docs.oracle.com/error-help/db/ora-02291/>

Commit complete.

6. When I try to insert the following entry into Employee with an invalid age of 17 (which is less than 18), I get this result:

```
insert into Employee values (10, 'Some Employee', 17, 1)
Error report -
ORA-02290: check constraint (APETER40.SYS_C0011664803) violated
```

<https://docs.oracle.com/error-help/db/ora-02290/>

More Details :

<https://docs.oracle.com/error-help/db/ora-02290/>

7. update Employee set Eage=70 where EID=1;

	EID	ENAME	EAGE	EDEPTID
1	1	Sachin Tendulkar	70	1

8. update Employee set Eage = Eage + 5;

	EID	ENAME	EAGE	EDEPTID
1	1	Sachin Tendulkar	75	1
2	2	Virat Kohli	50	2
3	3	Jasprit Bumrah	30	2
4	4	Rohit Sharma	35	3
5	5	Ravindra Jadeja	34	1

9. When I try to update the age of the following entry in Employee to an invalid age of 104 (which is above 99), I get this result:

```
update Employee set Eage = 104 where Eid = 4
Error report -
ORA-02290: check constraint (APETER40.SYS_C0011664803) violated
```

<https://docs.oracle.com/error-help/db/ora-02290/>

More Details :

<https://docs.oracle.com/error-help/db/ora-02290/>

10. delete Employee where EID = 5;

	EID	ENAME	EAGE	EDEPTID
1	1	Sachin Tendulkar	75	1
2	2	Virat Kohli	50	2
3	3	Jasprit Bumrah	30	2
4	4	Rohit Sharma	35	3

11. When I try to delete a row from Department while there are still entries in Employee with foreign keys attached to that entry, I get this error:

```
delete Department where DID = 1
```

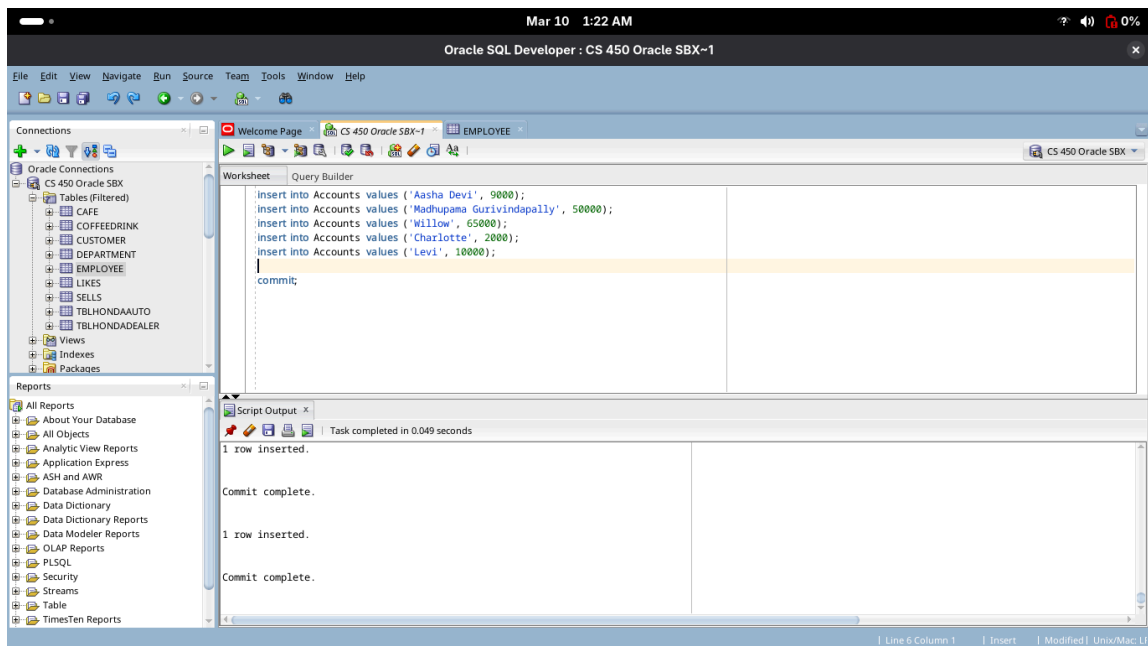
Error report -

```
ORA-02292: integrity constraint (APETER40.ED_ID) violated - child record found
```

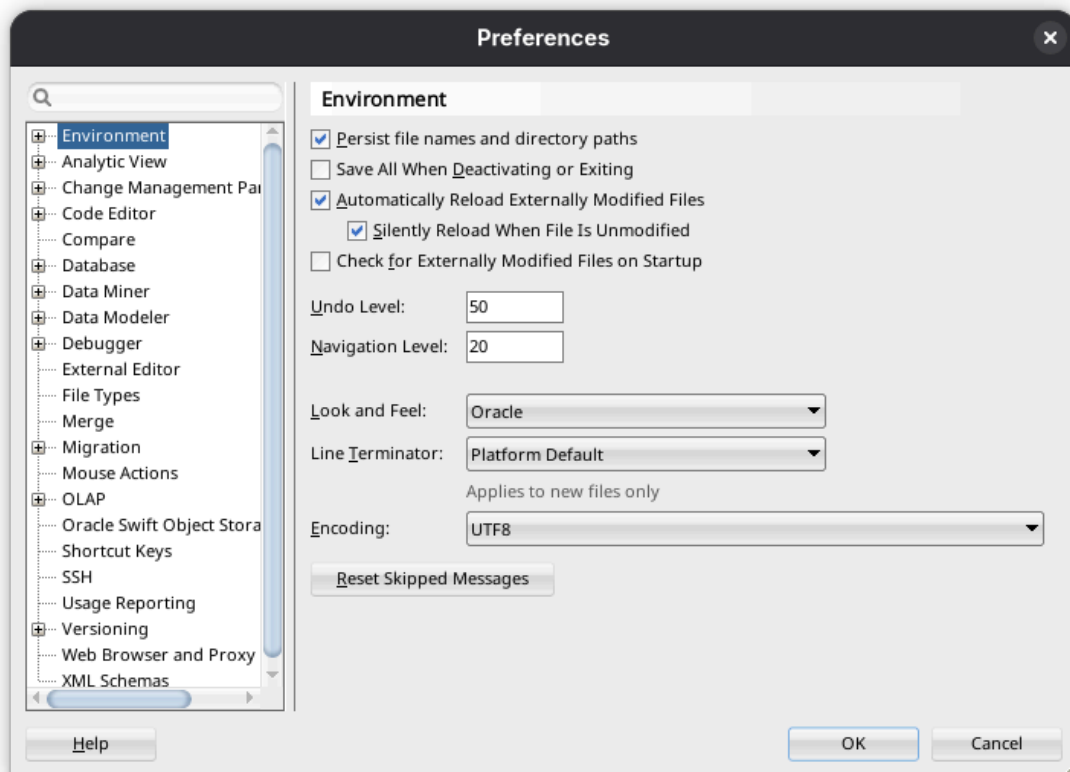
Part 2: Benefit of DBMS - Revisit Concurrency Issue

1. insert into Accounts values ('Aasha Devi', 9000);
insert into Accounts values ('Madhupama Gurivindapally', 50000);
insert into Accounts values ('Willow', 65000);
insert into Accounts values ('Charlotte', 2000);
insert into Accounts values ('Levi', 10000);

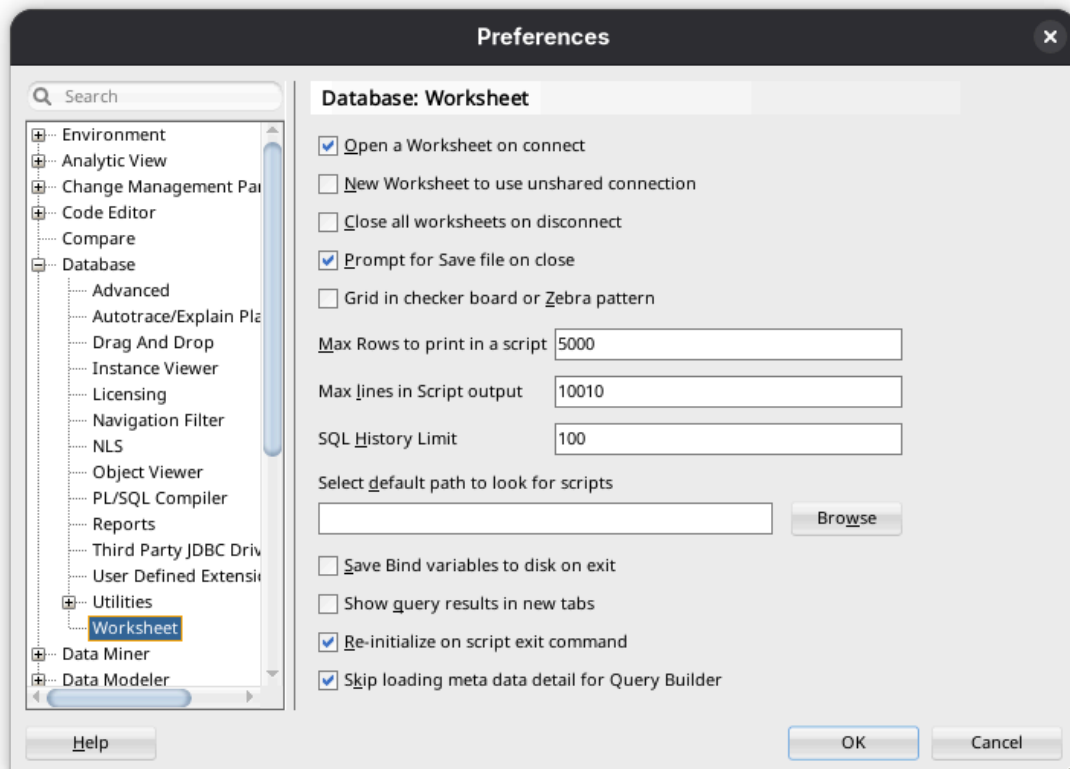
commit;



2.



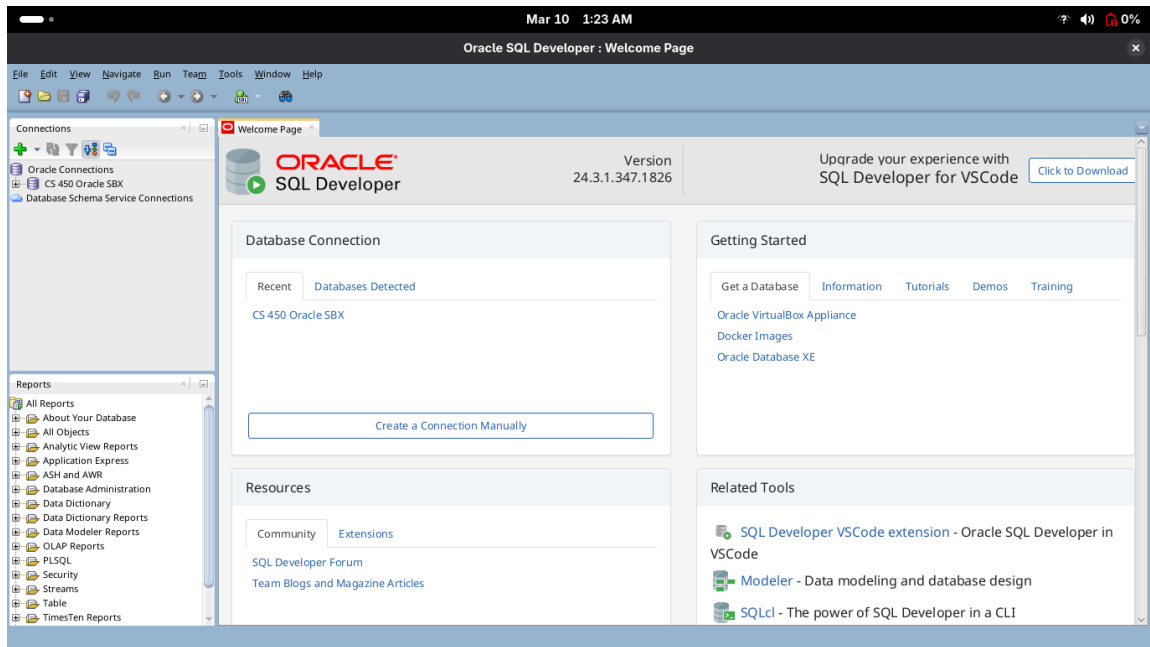
3.



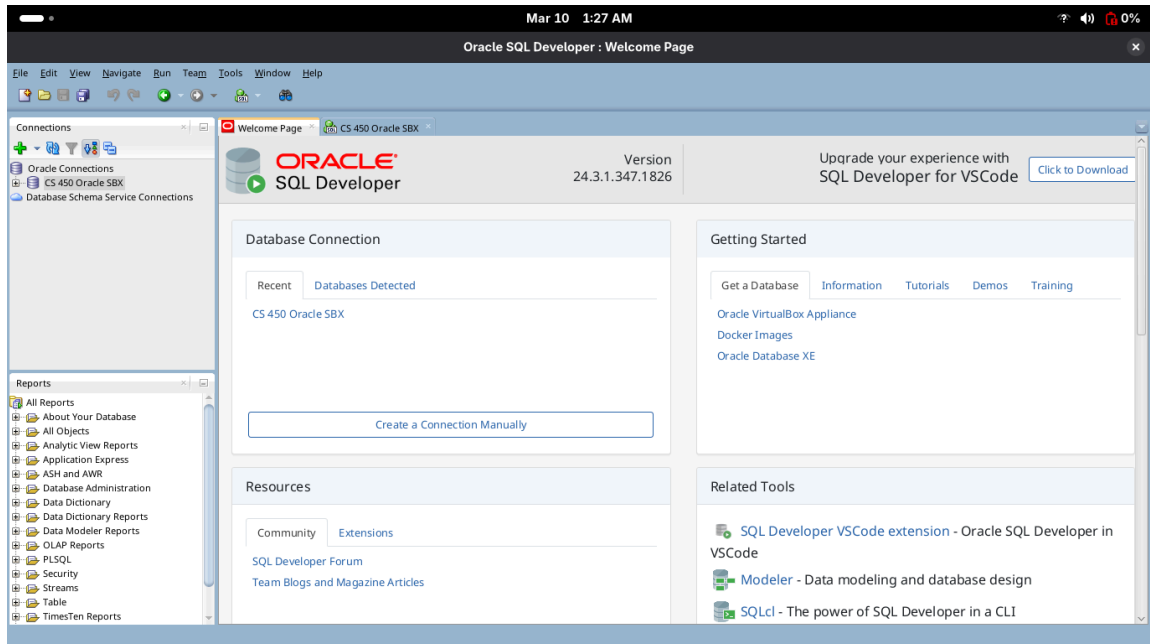
4.

5.

☒ New Worksheet to use unshared connection



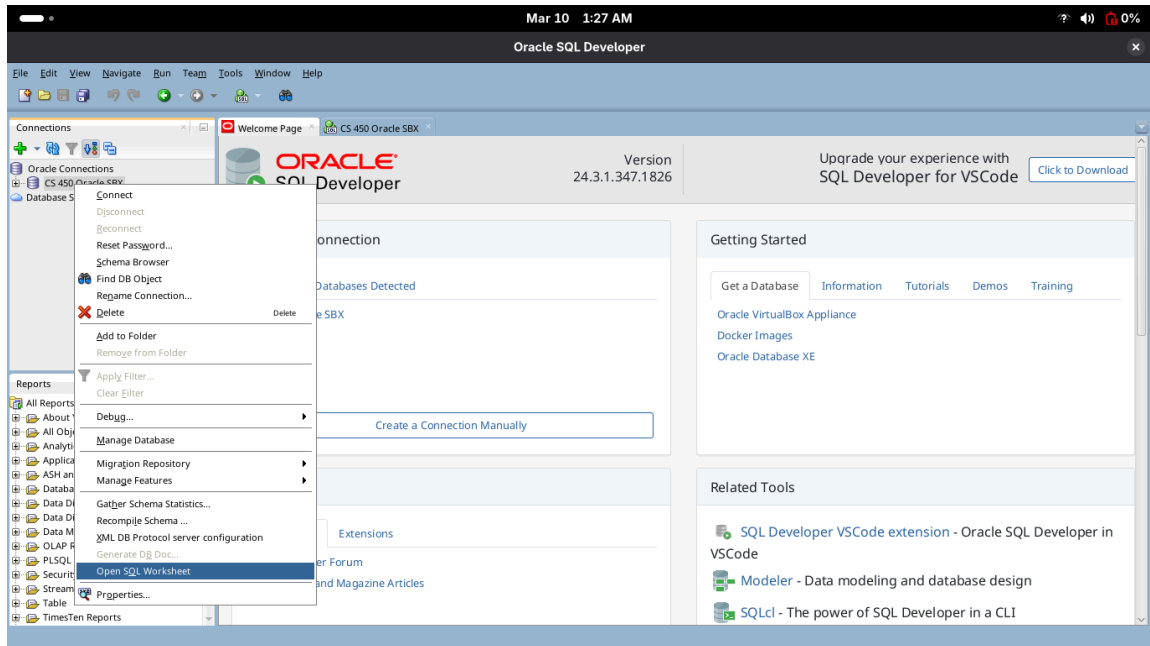
6.



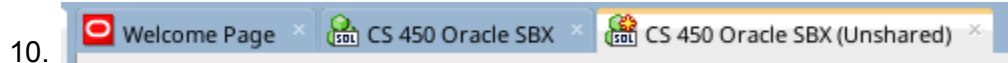
7.

8.





9.



10.

```

BALANCE
-----
          50

1 row updated.

```

11. From updating unshared worksheet:

12. No matter what I do, when running one worksheet after running another one before it, the one that runs second always stalls forever.